LeepTalent Level 3 End-point Assessment

Data Technician

SAMPLE

Assessment Method:

Scenario Demonstrations with Questioning

90 minutes (for Scenario 1 and 2)

Scenario Demonstration 1 – Data Gathering and Validation

45 minutes

You are employed as a data technician by a regional healthcare provider. The provider is reviewing patient admissions across its five hospitals and has asked you to prepare a single, validated data set that combines the admissions records with the hospital reference information, ready for analysis.

You have been provided with the following data sources: **Admissions.csv** (patient admission records), **Hospitals.xlsx** (hospital reference information) and the **BMI Classification** table in **Annex A**. Using a suitable data analysis tool of your choice, carry out the following:

Task 1 — Import the data sources

- Import **Admissions.csv** and **Hospitals.xlsx** into your chosen tool.
- Inspect the structure of both sources to confirm they have loaded correctly.

Task 2 — Identify and correct data quality issues

- Check both sources for data entry errors (typos, impossible values), inconsistent formats or naming, and missing or null values.
- Pay particular attention to the **lengthofstay** and **pulse** fields.

Task 3 — Merge the data sources

- Join the admissions records to the hospital information using the common identifier (**facid**).
- Ensure each admission now includes the correct hospital **Name**, and validate that all admissions have matched correctly.

Task 4 — Classify BMI

- Add a new column that classifies each patient's **bmi** into the categories shown in **Annex A**.

Your findings and any actions taken should be documented so that they are available for the team undertaking the analysis. Save the prepared data set with a clear name (for example, **Admissions_Cleaned.xlsx**).

Scenario Demonstration 2 – Data Analysis

45 minutes

Using the cleaned and combined data set from Scenario Demonstration 1, analyse how **length of stay** varies across different factors and present your findings in a format suitable for presentation to senior management.

Task 1 — Length of stay by hospital

- Group the data by hospital **Name** and calculate the average, minimum and maximum **lengthofstay** for each.
- Present the results as a summary table and a bar chart.

Task 2 — Length of stay by BMI category

- Using the BMI classification from Scenario 1, calculate the average **lengthofstay** for each BMI group.
- Present the results as a table and a bar or column chart.

Task 3 — Length of stay by gender

- Group patients by **gender** and calculate the average **lengthofstay** for each.
- Optionally break this down further by BMI or hospital.

Task 4 — Other variables

- Explore any relationships between **lengthofstay** and other numerical fields such as **pulse** and **respiration**, using correlation, scatter plots or trend lines.

Task 5 — Summary and presentation

- Produce a short summary of your findings, including key insights, any outliers or surprising patterns, and possible explanations or next steps.

Annex A – BMI Classification

This table shows how each patient's BMI should be classified in Scenario 1, Task 4.

BMI range	Category
Less than 18.5	Underweight
18.5 to 24.9	Healthy weight
25 to 29.9	Overweight
30 to 39.9	Obese
40 or above	Severely obese